

ORIGINAL RESEARCH

Evaluating evidential strength of statistically nonsignificant meta-analyses with likelihood ratios: a meta-epidemiological study

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Abstract

Objectives: This study aimed to determine the proportion of statistically nonsignificant meta-analyses that provide evidence favoring the null hypothesis of no difference, evidence favoring the alternative hypothesis, or inconclusive findings using likelihood ratios.

Study Design and Setting: In this meta-epidemiological study, Cochrane systematic reviews published in 2025 were screened. Eligible meta-analyses pooled data from two or more primary studies, reported a pooled effect estimate with a 95% CI including the value of no difference, and had a *P* value between 0.05 and 0.20. Meta-analyses using Hartung-Knapp/Sidik-Jonkman were excluded. In the primary analysis, likelihood ratios compared the null hypothesis against the observed pooled effect. In the secondary analysis, likelihood ratios were calculated against prespecified small, medium, and large effect thresholds; meta-analyses reporting weighted mean differences were excluded as no agreed-upon thresholds exist for these outcomes.

Results: A total of 936 meta-analyses from 172 Cochrane reviews were eligible. In the primary analysis, 693 (74%) provided weak evidence favoring the alternative hypothesis, and 243 (26%) were inconclusive; none favored the null hypothesis. All meta-analyses with *P* values between .05 and .10 showed weak evidence favoring the alternative, while those with *P* values between .11 and .20 were split between weak evidence favoring the alternative (55%) and inconclusive (45%) findings. In the secondary analysis of 732 meta-analyses, the proportion supporting the null hypothesis increased from 20% for small to 55% for medium to 75% for large effect thresholds, while inconclusive findings declined correspondingly.

Conclusion: Most nonsignificant meta-analyses provide weak evidence favoring the observed treatment effect rather than supporting no difference, but evidence increasingly favors no difference as prespecified effect thresholds increase in magnitude. Likelihood ratios enable guideline developers and clinicians to distinguish evidence of no difference from inconclusive findings and support more informed clinical recommendations and better-targeted research prioritization. © 2026 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

Keywords: Evidence synthesis; Evidence-based practice; Epidemiologic methods; Likelihood functions; Scholarly communication; Statistical data interpretation

1. Introduction

Robust meta-analyses based on well-conducted randomized clinical trials provide gold-standard evidence for clinical practice guidelines [1,2]. However, when such meta-analyses yield nonsignificant results, distinguishing evidence of no difference from inconclusive evidence or evidence showing a signal of a clinically relevant effect despite the lack of statistical significance between two interventions has important implications. This distinction influences certainty of evidence assessments for individual outcomes and contributes to the broader weighing of benefits and harms across multiple outcomes in guideline development [3]. Nonsignificant

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Plain Language Summary

When research combines results from multiple studies (called meta-analyses) and finds no statistically significant difference between treatments, researchers often interpret this to mean “there is no difference between the treatments.” However, this can be a mistake. Sometimes the data truly show the treatments work the same, but other times the data are simply unclear, and we just do not have enough information to tell either way. We used a statistical method called likelihood ratios to determine whether meta-analyses supported “no difference” or whether the results were simply unclear or favored a difference between the treatments. By examining 936 nonsignificant meta-analyses, we found that most results (74%) provided weak evidence that there is a difference between treatments, not that treatments work the same. However, when we compared results to what would count as clinically important differences, the picture changed. For small important differences, most results were unclear, while for large important differences, 75% supported “no difference.” This matters because misinterpreting unclear results as showing “no difference” can affect medical guidelines and patient care. It can also lead to wasted research by repeating studies when evidence already supports no difference or by stopping research when results are simply unclear and more data are needed. When researchers analyze combined results from studies, they should specify in advance what size of treatment difference would matter to patients, then use likelihood ratios to clearly state whether their results support no difference, are unclear, or suggest a treatment effect. This will help clinicians make better treatment decisions and researchers prioritize which questions still need more study.

pooled effect estimates are often misinterpreted as demonstrating “no difference” [4]. Such interpretations conflate two fundamentally distinct possibilities of the data genuinely supporting the null hypothesis of no difference or that the data are inconclusive [5,6], meaning they support neither the null nor alternative hypotheses. This distinction between “evidence of no difference” and “no evidence of difference” has important clinical implications [7].

Current statistical guidance addresses this interpretive challenge through P value thresholds and examination of confidence intervals (CIs). P values between .05 and .10 are characterized as providing “weak evidence” against the null hypothesis, while those above 0.10 provide “little to no evidence” [8]. CIs can indicate whether clinically meaningful values are excluded but typically encompass both meaningful and negligible effect sizes [9]. However, neither approach can directly evaluate evidence supporting alternative hypotheses of clinically meaningful differences without prespecified effect size thresholds.

The likelihood ratio quantifies the relative support for competing hypotheses by comparing how well each predicts the observed data [10]. Originally proposed for primary studies and subsequently adapted for application in meta-analysis [11], likelihood ratios can directly compare the null hypothesis against a prespecified alternative hypothesis without requiring prior probability distributions [12,13]. This allows researchers to determine whether the observed data are more compatible with no difference or with a clinically meaningful difference. For example, if a meta-analysis shows a risk ratio of 0.85 with a wide CI, the likelihood ratio quantifies whether these data better support “there is no difference” (ie, the null hypothesis) vs “there is a 15% reduction in risk” (ie, the alternative hypothesis).

Although the likelihood ratio has been investigated at scale in randomized clinical trials [14–17], its application in meta-analysis has been limited to illustrative examples in a small number of studies [18,19]. This study sought to fill this gap in knowledge by systematically evaluating the strength of evidence in nonsignificant meta-analyses from Cochrane systematic reviews with P values between .05 and .20.

2. Methods

2.1. Study design and registration

This meta-epidemiological study is reported in accordance with the modified Preferred Reporting Items for Systematic Reviews and Meta-analyses statement for meta-epidemiological methodology research [20]. The study protocol was prospectively registered with the Open Science Framework (<https://doi.org/10.17605/OSF.IO/J9RCY>).

2.2. Eligibility criteria

Meta-analyses that pooled data from two or more primary studies and reported pooled effect estimates with 95% CIs that included the null value (1 for relative measures and 0 for absolute measures) were included. Reviews of diagnostic test accuracy, qualitative reviews, reviews without an analysis section, meta-analyses that did not report individual study-level data, and meta-analyses that used Hartung-Knapp/Sidik-Jonkman or other t distribution–based CI methods were excluded. The latter exclusion was necessary because the likelihood ratio framework requires z -statistics derived from Wald-type standard errors, and including t -based methods would create inconsistency between the selection criteria (P value =

What is new?**Key findings**

- Evidence interpretation shifted from favoring an effect to favoring no difference depending on whether observed or prespecified effect thresholds were used as alternative hypotheses.
- Using observed effects as alternative hypotheses is discouraged due to circular reasoning.

What this adds to what was known?

- This study is the first large-scale empirical demonstration that likelihood ratios can systematically distinguish evidence of no difference from inconclusive findings in meta-analysis.

What is the implication and what should change now?

- Clinical guideline developers can use this framework to make more nuanced recommendations, particularly for safety outcomes where distinguishing no effect from inconclusive evidence is important.
- Research prioritization should differentiate nonsignificant results showing strong evidence for no difference from those that are inconclusive.

.05–.20) and the evidence quantification. There was no limit on the number of eligible meta-analyses from a single review and no restrictions on meta-analysis type (pairwise or network meta-analysis).

2.3. Information sources and search strategy

The Cochrane Library was searched for all systematic reviews published between January 1 and December 31, 2025. There were no restrictions on clinical areas or intervention type.

2.4. Study selection

One reviewer screened all reviews and extracted data from eligible meta-analyses. After selecting all meta-analyses with CIs that included the null value, pooled effect estimates were converted to *P* values using the formula proposed by Altman and Bland [21]. Analysis was restricted to meta-analyses with *P* values between .05 and .20 to encompass the thresholds proposed for weak evidence (0.05–0.10) and little to no evidence (>0.10) against the null hypothesis [8] while maintaining analytical feasibility.

Each eligible meta-analysis was replicated using the exact statistical methods reported by the original authors to obtain accurate *z*-statistics (the pooled effect estimate divided by its standard error) and to avoid potential inaccuracies from rounding of published estimates. Replication accuracy was verified by confirming that replicated pooled effect estimates matched those reported in the original reviews within two decimal places.

2.5. Data collection process

Data extraction captured the clinical topic and number of included meta-analyses for each eligible review along with methodological details for each meta-analysis, including the statistical model, pooling method, summary measure, and methods for estimating CIs and between-study heterogeneity. Each outcome was categorized as measuring efficacy or safety based on the authors' descriptions or, when unavailable, clinical interpretation. Individual study-level data included the number of events and total participants in each treatment arm for binary outcomes, and means, SDs, and sample sizes for each arm for continuous outcomes.

2.6. Synthesis of results

Likelihood ratios were calculated by first computing the natural logarithm of the likelihood ratio (termed "support") [11], comparing prespecified alternative hypotheses against the null hypothesis of no difference, then exponentiating the support value to obtain the final likelihood ratio value.

The primary analysis used observed pooled effects as the alternative hypothesis. The secondary analysis used effect size thresholds representing small, medium, and large effects, oriented toward the favorable direction of effect. For binary outcomes reported as odds ratios or risk ratios, mathematically symmetric thresholds were used of small (0.82, 1.22), medium (0.54, 1.86), and large (0.33, 3.00), which maintain equal distances from the null hypothesis (value of 1.0) on the log scale [22]. For risk differences, values of -0.05 (small), -0.10 (medium), and -0.15 (large) were used [23]. For standardized mean differences, Cohen's *d* thresholds of ± 0.2 (small), ± 0.5 (medium), and ± 0.8 (large) were used [24]. Weighted mean differences were excluded from the secondary analysis due to the absence of agreed-upon thresholds across clinically heterogeneous outcome scales.

Likelihood ratios were interpreted using the framework proposed by Goodman [11] with minor modifications. In this framework, values greater than 1 indicate evidence favoring the alternative hypothesis, and values less than 1 indicate evidence favoring the null hypothesis. Likelihood ratios of 2.7, 7.4, 20, and 55 (and their reciprocals) represent weak, moderate, strong, and extremely strong

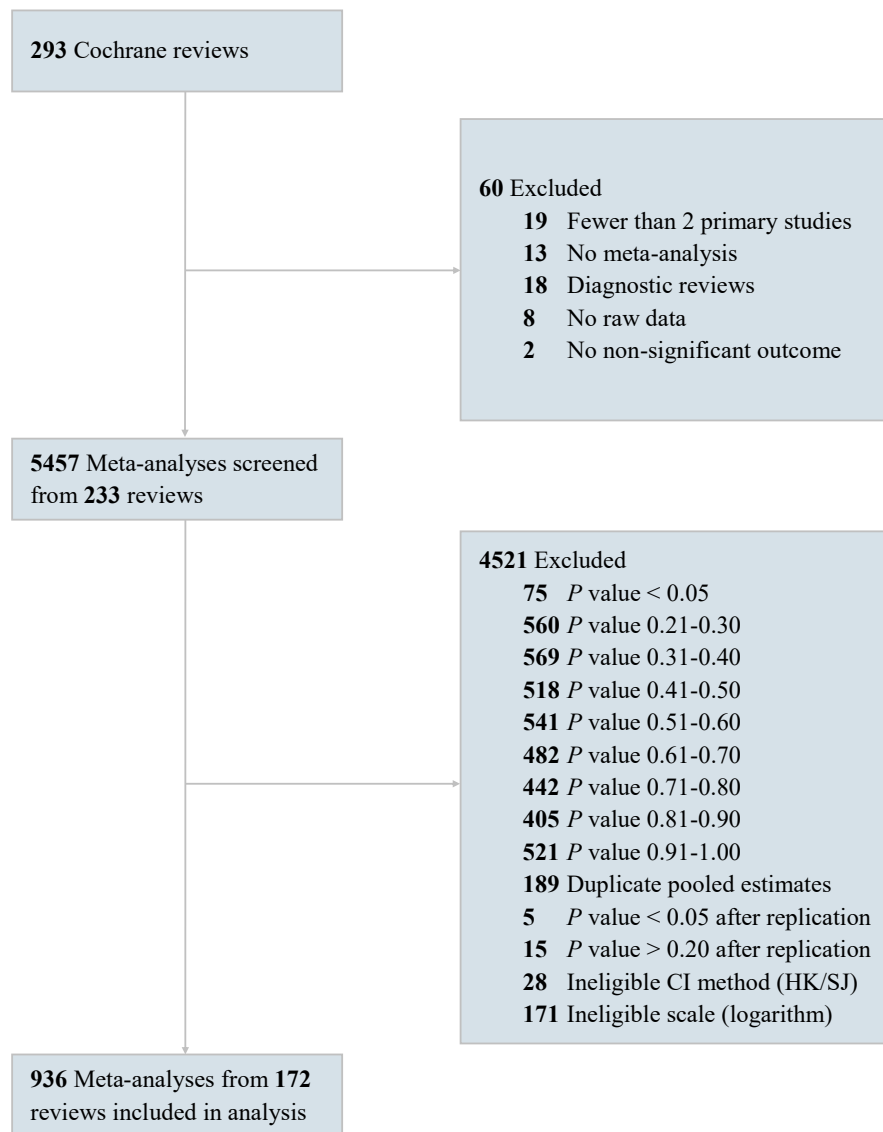


Figure. Study selection flow diagram. HK, Hartung-Knapp; SJ, Sidik-Jonkman.

evidence, respectively. Likelihood ratios between 0.37 and 2.7 were classified as inconclusive.

The distribution of likelihood ratios across all eligible meta-analyses was reported using descriptive statistics (median and first and third quartiles) and stratified analyses by outcome type (binary vs continuous). Associations between likelihood ratios and P values were examined using Spearman's rank correlation test [25]. All analyses were conducted in R (version 4.5.1; R Core Team, Vienna, Austria) via RStudio (version 2025.9.2.418; Posit Software, Boston, MA), and the meta-R package [26] was used to replicate meta-analyses and obtain z -statistics for likelihood ratio calculations.

3. Results

Of 5457 meta-analyses from 233 Cochrane reviews, 936 from 172 reviews met eligibility criteria (Fig). All 936 meta-analyses were included in the primary analysis. The secondary analysis included 732 meta-analyses after excluding 204 with weighted mean differences. Characteristics of included reviews and meta-analyses are available in Table 1 and Supplementary Table 1.

3.1. Primary analysis

The median likelihood ratio was 3.4 (Q1–Q3, 2.7–4.6) across 936 meta-analyses. Most results ($n = 693$, 74.0%)

Table 1. Characteristics of included Cochrane reviews and meta-analyses

Characteristic	Values
Clinical topic <i>n/N</i> (%)	
Anesthesiology	3/172 (1.7)
Cardiology	11/172 (6.4)
Critical care	2/172 (1.2)
Dermatology	1/172 (0.6)
Emergency medicine	1/172 (0.6)
Endocrinology	6/172 (3.5)
Gastroenterology	7/172 (4.1)
Geriatrics	1/172 (0.6)
Hematology	3/172 (1.7)
Infectious diseases	14/172 (8.1)
Nephrology	4/172 (2.3)
Neurology	18/172 (10.5)
Nutrition	1/172 (0.6)
Obstetrics and gynecology	22/172 (12.8)
Oncology	8/172 (4.6)
Ophthalmology	1/172 (0.6)
Orthopedics	7/172 (4.1)
Palliative care	1/172 (0.6)
Pediatrics	16/172 (9.3)
Psychiatry	20/172 (11.6)
Pulmonology	3/172 (1.7)
Rheumatology	4/172 (2.3)
Surgery	13/172 (7.6)
Urology	5/172 (2.9)
Number of statistically nonsignificant outcomes per review	
Median (Q1–Q3)	3.0 (2.0–7.0)
Mean (SD)	5.4 (6.2)
Number of primary studies per meta-analysis	
Median (Q1–Q3)	3.0 (2.0–5.0)
Mean (SD)	4.7 (4.4)
Sample size per meta-analysis	
Median (Q1–Q3)	504.5 (227.5–1514.8)
Mean (SD)	2514.1 (6092.9)
Outcome type <i>n/N</i> (%)	
Efficacy	579/936 (62.0)
Safety	357/936 (38.0)

Data represent 936 statistically nonsignificant meta-analyses (P value = .05–.20) from 172 Cochrane systematic reviews published in 2025. Sample size indicates total participants across all primary studies within each meta-analysis. Data are presented as *n/N* (%) or median (Q1–Q3), with mean (SD) also reported for selected characteristics.

provided weak evidence favoring the alternative hypothesis, and the remainder were inconclusive. All meta-analyses with P values between .05 and .10 showed weak evidence favoring the alternative ($n = 401$), whereas those with P values between .11 and .20 were divided between weak evidence favoring the alternative ($n = 292$, 54.6%) and inconclusive findings ($n = 243$, 45.4%; Table 2).

Likelihood ratios and P values showed a negative correlation ($\rho = -1.000$; $P < .001$).

3.2. Secondary analysis

Evidence patterns varied substantially by effect size threshold across 732 meta-analyses. The median likelihood ratios were 1.7 (Q1–Q3, 0.5–2.7) for small effects, 0.2 (Q1–Q3, 1.9×10^{-3} –2.2) for medium effects, and 1.4×10^{-3} (Q1–Q3, 2.1×10^{-10} –0.4) for large effects. As the threshold magnitude increased, evidence increasingly favored the null hypothesis over the alternative. This pattern held across all outcome types and P value ranges (Table 3). Correlations between likelihood ratios and P values were weak and negative for small effects ($\rho = -0.307$; $P < .001$), weak and negative for medium effects ($\rho = -0.118$; $P = .001$), and negligible for large effects ($\rho = 0.002$; $P = .956$).

Table 4 provides illustrative examples from the secondary analysis, selected to represent the range of evidence categories observed. A worked example of likelihood ratio application and interpretation is provided in the supplementary material, with the calculation steps presented in Supplementary Table 2.

4. Discussion

The findings of this study show that the interpretation of nonsignificant meta-analyses depends fundamentally on the prespecified alternative hypothesis. Using observed effects as alternatives, most meta-analyses provided weak evidence for the observed treatment effect. This pattern challenges the common interpretation of nonsignificant results as showing “no difference”; when data are evaluated against what was observed, they typically favor that effect over no difference, albeit weakly. In contrast, when evaluated against prespecified clinically important thresholds, the picture changed dramatically. Most results remained inconclusive for small effects, while evidence increasingly favored no difference for medium and large effects. This finding contradicts a specific form of misinterpretation, particularly common for results with P values between .05 and .10, a range often described as showing “trends toward significance” or “approaching significance” [34]. The comparison between the primary and secondary analyses highlights that likelihood ratio evaluations are conceptually more meaningful when the alternative hypothesis is prespecified rather than derived from the observed data, as the latter introduces circularity into the interpretation.

These empirical findings are consistent with theoretical predictions about the interpretation of nonsignificant results. Prior work demonstrated that likelihood ratios quantify evidence in individual trials more interpretably than P values alone, with apparently promising results often

Table 2. Distribution of likelihood ratios

Analysis category	Evidence category			
	Favors the null hypothesis	Inconclusive	Favors the alternative hypothesis	
	Weak to extremely strong for null (LR ≤ 0.37)	No evidence for either hypothesis over the other (LR > 0.37 to < 2.7)	Weak for alternative (LR 2.7 to < 7.4)	Moderate to extremely strong for alternative (LR ≥ 7.4)
All meta-analyses ($n = 936$)	0 (0)	243 (26.0)	693 (74.0)	0 (0)
Binary outcomes ($n = 617$)	0 (0)	166 (26.9)	451 (73.1)	0 (0)
Continuous outcomes ($n = 319$)	0 (0)	77 (24.1)	242 (75.9)	0 (0)
P value = .05–.10 ($n = 401$)	0 (0)	0 (0)	401 (100.0)	0 (0)
P value = .11–.20 ($n = 535$)	0 (0)	243 (45.4)	292 (54.6)	0 (0)

LR, likelihood ratio.

Likelihood ratios quantify evidence for the alternative hypothesis relative to the null hypothesis of no difference. Alternative hypotheses represent the observed effects. Data are presented as n (%) for likelihood ratio categories.

yielding only modest likelihood ratios favoring alternative over the null hypotheses [35]. Other investigators showed mathematically that P values near the conventional alpha threshold of .05 do not reliably predict future significance with additional data [36].

4.1. Circular reasoning with observed effects as alternative hypotheses

Specifying the alternative hypothesis as the observed effect size introduces circular reasoning because the same data both estimate the effect and define what constitutes evidence for that effect. This was evident in our findings where no meta-analysis evaluated against its own observed effect showed evidence favoring the null hypothesis. In contrast, evaluations against prespecified thresholds yielded diverse evidence patterns. This circularity, described as the Texas sharpshooter fallacy [35,37], demonstrates that prespecified alternative hypotheses are essential for valid evidence interpretation.

4.2. Implications for evidence synthesis

The requirement to prespecify meaningful alternatives formalizes a practice already recommended by Grading of Recommendations Assessment, Development and Evaluation (GRADE), where systematic reviewers predetermine decision thresholds for assessing imprecision [38]. However, while GRADE examines whether CIs cross these thresholds, likelihood ratios quantify whether imprecise estimates favor no difference, remain inconclusive, or provide evidence for an effect of interest. Meta-analysts could use this framework to support more nuanced guideline recommendations, particularly for safety outcomes. It could also inform research prioritization by identifying

which negative results close research questions vs which signal knowledge gaps and reduce research waste from unnecessary replication of studies showing strong evidence for no difference. Likelihood ratio thresholds serve as benchmarks to guide verbal characterization of the strength of evidence rather than as rigid cut-offs, as the scale of statistical evidence is continuous and not discrete [13,39].

4.3. Limitations

This study has several shortcomings. First, a single reviewer conducted the screening and data extraction, which may have introduced errors or inconsistencies. However, the replication of published meta-analyses with verification against reported results within two decimal places provided a quality check on the accuracy of the data extraction. Second, the analysis was restricted to meta-analyses with P values between .05 and .20 to focus on results most susceptible to misinterpretation while maintaining analytical feasibility. This exclusion of meta-analyses with P values above .20 limits the generalizability of our findings to all nonsignificant results, although results above this value are less likely to be misinterpreted as showing a clinically meaningful difference. Third, multiple meta-analyses from individual Cochrane reviews were included, which may not be statistically independent. Nevertheless, our descriptive analysis does not rely on statistical inference across meta-analyses, which makes this dependency less problematic. Fourth, while likelihood ratios and the direction of evidence are calculated objectively from the data, the thresholds used to categorize them into evidence strength categories follow published conventions

Table 3. Distribution of likelihood ratios by alternative hypothesis effect size and outcome type

Evidence category	Small effect	Medium effect	Large effect
All meta-analyses (<i>N</i> = 732)			
Extremely strong for null (LR < 0.018)	45 (6.2)	235 (32.1)	441 (60.3)
Strong for null (LR 0.018–0.05)	12 (1.6)	48 (6.6)	31 (4.2)
Moderate for null (LR 0.05–0.14)	31 (4.2)	53 (7.2)	36 (4.9)
Weak for null (LR 0.14–0.37)	61 (8.3)	62 (8.5)	37 (5.1)
Inconclusive (LR 0.37–2.7)	391 (53.4)	209 (28.5)	118 (16.1)
Weak for alternative (LR 2.7–7.4)	192 (26.2)	125 (17.1)	69 (9.4)
Binary (<i>n</i> = 617)			
Extremely strong for null (LR < 0.018)	39 (6.3)	212 (34.4)	380 (61.6)
Strong for null (LR 0.018–0.05)	10 (1.6)	39 (6.3)	26 (4.2)
Moderate for null (LR 0.05–0.14)	28 (4.5)	47 (7.6)	28 (4.5)
Weak for null (LR 0.14–0.37)	51 (8.3)	50 (8.1)	30 (4.9)
Inconclusive (LR 0.37–2.7)	348 (56.4)	171 (27.7)	96 (15.6)
Weak for alternative (LR 2.7–7.4)	141 (22.9)	98 (15.9)	57 (9.2)
Continuous (<i>n</i> = 115)			
Extremely strong for null (LR < 0.018)	6 (5.2)	23 (20.0)	61 (53.0)
Strong for null (LR 0.018–0.05)	2 (1.7)	9 (7.8)	5 (4.4)
Moderate for null (LR 0.05–0.14)	3 (2.6)	6 (5.2)	8 (6.9)
Weak for null (LR 0.14–0.37)	10 (8.7)	12 (10.4)	7 (6.1)
Inconclusive (LR 0.37–2.7)	44 (38.3)	38 (33.0)	22 (19.1)
Weak for alternative (LR 2.7–7.4)	50 (43.5)	27 (23.5)	12 (10.4)
<i>P</i> value .050–.10 (<i>n</i> = 305)			
Extremely strong for null (LR < 0.018)	17 (5.6)	101 (33.1)	192 (62.9)
Strong for null (LR 0.018–0.05)	5 (1.6)	20 (6.6)	12 (3.9)
Moderate for null (LR 0.05–0.14)	11 (3.6)	22 (7.2)	13 (4.3)
Weak for null (LR 0.14–0.37)	22 (7.2)	21 (6.9)	8 (2.6)
Inconclusive (LR 0.37–2.7)	111 (36.4)	53 (17.4)	30 (9.8)
Weak for alternative (LR 2.7–7.4)	139 (45.6)	88 (28.8)	50 (16.4)
<i>P</i> value .11–.20 (<i>n</i> = 427)			
Extremely strong for null (LR < 0.018)	28 (6.6)	134 (31.4)	249 (58.3)
Strong for null (LR 0.018–0.05)	7 (1.6)	28 (6.6)	19 (4.4)
Moderate for null (LR 0.05–0.14)	20 (4.7)	31 (7.2)	23 (5.4)
Weak for null (LR 0.14–0.37)	39 (9.1)	41 (9.6)	29 (6.8)
Inconclusive (LR 0.37–2.7)	281 (65.8)	156 (36.5)	88 (20.6)
Weak for alternative (LR 2.7–7.4)	52 (12.2)	37 (8.7)	19 (4.4)

LR, likelihood ratio.

Likelihood ratios quantify evidence for the alternative hypothesis relative to the null hypothesis of no difference. Each meta-analysis was evaluated against three prespecified alternative hypothesis thresholds. Alternative effect sizes for binary outcomes using relative measures: 0.82 or 1.22 (small), 0.54 or 1.86 (medium), 0.33 or 3.00 (large); absolute measures -0.05 (small), -0.10 (medium), and -0.15 (large); for continuous outcomes using standardized mean differences: ± 0.2 (small), ± 0.5 (medium), and ± 0.8 (large). Data are presented as *n* (%) for likelihood ratio categories.

[10,11], although they remain subjective. Fifth, as the likelihood ratio assumes a normal distribution and is estimated via the *z*-statistic [35], the exclusion of meta-analyses using Hartung-Knapp/Sidik-Jonkman methods may limit applicability to some modern meta-analytic practices, given that the choice of heterogeneity

estimator and interval method can affect interpretation in meta-analyses. Finally, this study was limited to Cochrane systematic reviews published in 2025, which represent high-quality evidence synthesis but may not reflect the broader landscape of meta-analyses in the published literature.

Table 4. Statistical evidence from selected Cochrane meta-analyses as measured by likelihood ratios.

Cochrane review excerpt	Likelihood ratio interpretation
“We evaluated mortality at 90 days...and found...no significant impacts by transfusion strategy, albeit with some variations between clinical contexts.” [27]	Evidence regarding the effect of restrictive vs liberal transfusion thresholds on 90-d mortality in adults is extremely strong in favor of the null hypothesis of no difference (LR = 0.0009).
“Adverse events (RR 1.49, 95% CI 0.82 to 2.7; $P = 0.2$, $I^2 = 0\%$; 4 studies, 1166 participants...) and infection (RR 2.53, 95% CI 0.84 to 7.61; $P = 0.10$, $I^2 = 0\%$; 3 studies, 995 participants...) were not different between groups [comparing interrupted vs. uninterrupted anticoagulation in cardiac implantable electronic device surgery].” [28]	Evidence regarding the effect of interrupted vs uninterrupted anticoagulation on adverse events is weak in favor of no difference (LR = 0.35), while evidence regarding infection is inconclusive (LR = 0.52).
“There may be little to no difference between renin inhibitors and ARBs [angiotensin receptor blockers] in all-cause mortality (RR 0.35, 95% CI 0.07 to 1.64; 3 studies, 1932 participants; low-certainty evidence).” [29]	Evidence regarding the effect of renin inhibitors vs angiotensin receptor blockers on all-cause mortality is inconclusive (LR = 1.35).
“...there may be no reduction in 30-day mortality [when prophylactic transarterial embolization following successful endoscopic treatment is compared with endoscopic haemostasis alone] (OR 0.41, 95% CI 0.14 to 1.21; 2 studies, 346 participants; low-certainty evidence).” [30]	Evidence regarding the effect of prophylactic transarterial embolization vs endoscopic hemostasis on 30-d mortality is inconclusive (LR = 1.67).
“There was no difference [in somnolence] between antipsychotic switching and antipsychotic continuation groups (RR 0.46, 95% CI 0.17 to 1.24; 2 studies, 509 participants...).” [31]	Evidence regarding the effect of antipsychotic switching vs antipsychotic continuation on somnolence is inconclusive (LR = 1.69)
“Thrombolysis...when compared to placebo or standard care... likely results in little to no difference in the risk of hydrocephalus (18.3% versus 24.1%, RR 0.77, 95% CI 0.54 to 1.10; [6] studies, [341] participants; moderate-certainty evidence).” [32]	Evidence regarding the effect of thrombolysis vs control on the risk of hydrocephalus is weak in favor of the intervention (LR = 2.71).
“Craniotomy with haematoma evacuation...may reduce 30-day case fatality (RR 0.68, 95% CI 0.46 to 1.00; 3 studies, 676 participants; low-certainty evidence), but the pooled CIs of 30-day case fatality included the possibility of no effect.” [33]	Evidence regarding the effect of craniotomy with hematoma evacuation vs standard medical management alone on 30-d case fatality is weak in favor of the intervention (LR = 4.29).

LR, likelihood ratio; OR, odds ratio; RR, relative risk.

Likelihood ratios quantify evidence for the alternative hypothesis relative to the null hypothesis of no difference. Examples were selected to represent the range of evidence categories observed. Each excerpt is reproduced verbatim from the original Cochrane review, with ellipses indicating omitted text and square brackets indicating insertions for clarity. The small effect alternative hypothesis of 0.82 was applied to all relative measures.

5. Conclusion

The likelihood ratio framework provides clearer interpretation for nonsignificant findings than examining whether CIs cross the line of no difference. By distinguishing evidence favoring an effect from inconclusive results, likelihood ratios offer distinctions that P values or CIs cannot provide. Meta-analysts should consider prespecifying clinically important effect sizes and reporting likelihood ratios alongside pooled effect estimates. This approach enables decision-makers to determine whether results favor a meaningful effect, favor no difference, or remain inconclusive. Distinguishing inconclusive from evidence compatible with no difference would improve research prioritization, reduce waste from unnecessary research, and support clinical recommendations that appropriately reflect the strength and direction of available evidence. Future research should validate these findings in broader systematic review databases beyond Cochrane, evaluate the framework’s application to significant results,

and develop standardized guidance for selecting effect size thresholds across diverse clinical domains.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Claude to assist with the initial development of code. After using this tool, the authors reviewed, refined, and edited the code as needed and verified its validity against published literature. The authors take full responsibility for the content of the published article.

CRedit authorship contribution statement

Amin Sharifan: Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Funding acquisition,

Formal analysis, Data curation, Conceptualization. **Curtis Harrod:** Writing — review & editing, Supervision, Methodology, Conceptualization, Investigation, Validation. **Andreea Dobrescu:** Writing — review & editing, Supervision. **Gerald Gartlehner:** Writing — review & editing, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2026.112297>.

Data availability

Data will be made available on request.

References

- [1] Dechartres A, Altman DG, Trinquart L, Boutron I, Ravaud P. Association between analytic strategy and estimates of treatment outcomes in meta-analyses. *JAMA* 2014;312:623–30. <https://doi.org/10.1001/jama.2014.8166>.
- [2] Kanters S, Ford N, Druyts E, Thorlund K, Mills EJ, Bansback N. Use of network meta-analysis in clinical guidelines. *Bull World Health Organ* 2016;94:782–4. <https://doi.org/10.2471/BLT.16.174326>.
- [3] Hultcrantz M, Rind D, Akl EA, Treweek S, Mustafa RA, Iorio A, et al. The GRADE Working Group clarifies the construct of certainty of evidence. *J Clin Epidemiol* 2017;87:4–13. <https://doi.org/10.1016/j.jclinepi.2017.05.006>.
- [4] Uttley L, Weng Y, Falzon L. Yet another problem with systematic reviews: a living review update. *J Clin Epidemiol* 2025;177:111608. <https://doi.org/10.1016/j.jclinepi.2024.111608>.
- [5] Altman DG, Bland JM. Statistics notes: absence of evidence is not evidence of absence. *BMJ* 1995;311:485. <https://doi.org/10.1136/bmj.311.7003.485>.
- [6] Naylor CD. Meta-analysis and the meta-epidemiology of clinical research. *BMJ* 1997;315:617. <https://doi.org/10.1136/bmj.315.7109.617>.
- [7] LeLorier J, Grégoire G, Benhaddad A, Lapierre J, Derderian F. Discrepancies between meta-analyses and subsequent large randomized, controlled trials. *N Engl J Med* 1997;337:536–42. <https://doi.org/10.1056/NEJM199708213370806>.
- [8] Held L, Ott M. On *P* values and Bayes factors. *Annu Rev Stat Appl* 2018;5:393–419. <https://doi.org/10.1146/annurev-statistics-031017-100307>.
- [9] Hawkins AT, Samuels LR. Use of confidence intervals in interpreting nonstatistically significant results. *JAMA* 2021;326:2068–9. <https://doi.org/10.1001/jama.2021.16172>.
- [10] Goodman SN, Royall R. Evidence and scientific research. *Am J Public Health* 1988;78:1568–74. <https://doi.org/10.2105/AJPH.78.12.1568>.
- [11] Goodman SN. Meta-analysis and evidence. *Control Clin Trials* 1989; 10:188–204. [https://doi.org/10.1016/0197-2456\(89\)90030-5](https://doi.org/10.1016/0197-2456(89)90030-5).
- [12] Goodman SN. Values, hypothesis tests, and likelihood: implications for epidemiology of a neglected historical debate. *Am J Epidemiol* 1993; 137:485–96. <https://doi.org/10.1093/oxfordjournals.aje.a116700>.
- [13] Blume JD. Likelihood methods for measuring statistical evidence. *Stat Med* 2002;21:2563–99. <https://doi.org/10.1002/sim.1216>.
- [14] Perneger T, Gayet-Ageron A. Evidence of lack of treatment efficacy derived from statistically nonsignificant results of randomized clinical trials. *JAMA* 2023;329:2050–6. <https://doi.org/10.1001/jama.2023.8549>.
- [15] Sun H, Gershengorn HB, Mittel AM, Hua M. Using likelihood ratios to aid the interpretation of nonsignificant results in sepsis-related randomized controlled trials. *Ann Am Thorac Soc* 2026;23:313–7. <https://doi.org/10.1093/annalsats/aaof036>.
- [16] Noll D, Stapleton P, O'Callaghan M. Urological trials without significant results. Can we extract meaning from disappointment? *BJU Int* 2026. <https://doi.org/10.1111/bju.70140>.
- [17] Khan MS, Jamil A, Januzzi JL, Shakoor M, Bennett MM, Vanzyl JS, et al. POINT: statistical non-significance, likelihood ratio, and the interpretation of clinical trial evidence: insights from heart failure randomized trials. *J Card Fail* 2024;30:1629–32. <https://doi.org/10.1016/j.cardfail.2024.07.026>.
- [18] Dormuth CR, Filion KB, Platt RW. Likelihood ratio meta-analysis: new motivation and approach for an old method. *Contemp Clin Trials* 2016;47:259–65. <https://doi.org/10.1016/j.cct.2016.01.008>.
- [19] Thomas R, Daoud M. A comparison of weighted pooled results from a cochrane meta-analysis with likelihood ratio functions to determine strength of evidence. In: *Bulletin of the ISI 58th World Statistics Congress of the International Statistical Institute*. Dublin, Ireland: International Statistical Institute; 2011:6529–36. Available at: <https://isi-web.org/sites/default/files/import/files-2011/950591.pdf>. Accessed May 14, 2026.
- [20] Murad MH, Wang Z. Guidelines for reporting meta-epidemiological methodology research. *BMJ Evid Based Med* 2017;22:139. <https://doi.org/10.1136/ebmed-2017-110713>.
- [21] Altman DG, Bland JM. How to obtain the *P* value from a confidence interval. *BMJ* 2011;343:d2304. <https://doi.org/10.1136/bmj.d2304>.
- [22] Olivier J, May WL, Bell ML. Relative effect sizes for measures of risk. *Commun Stat Theor Methods* 2017;46:6774–81. <https://doi.org/10.1080/03610926.2015.1134575>.
- [23] Zeng L, Brignardello-Petersen R, Guyatt G. When applying GRADE, how do we decide the target of certainty of evidence rating? *Evid Based Ment Health* 2021;24:121–3. <https://doi.org/10.1136/ebmental-2020-300170>.
- [24] Cohen J. A power primer. *Psychol Bull* 1992;112:155–9. <https://doi.org/10.1037/0033-2909.112.1.155>.
- [25] Schober P, Boer C, Schwarte LA. Correlation coefficients: appropriate use and interpretation. *Anesth Analg* 2018;126:1763–8. <https://doi.org/10.1213/ANE.0000000000002864>.
- [26] Balduzzi S, Rücker G, Schwarzer G. How to perform a meta-analysis with R: a practical tutorial. *BMJ Ment Health* 2019;22:153–60. <https://doi.org/10.1136/ebmental-2019-300117>.
- [27] Carson JL, Stanworth SJ, Dennis JA, Fergusson DA, Pagano MB, Roubinian NH, et al. Transfusion thresholds and other strategies for guiding red blood cell transfusion. *Cochrane Database Syst Rev* 2025. <https://doi.org/10.1002/14651858.CD002042.pub6>.
- [28] Chen B, Phan M, Pasupuleti V, Roman YM, Hernandez AV. Interrupted vs uninterrupted anticoagulation for cardiac rhythm management device insertion. *Cochrane Database Syst Rev* 2025. <https://doi.org/10.1002/14651858.CD013816.pub2>.
- [29] Wang GM, Li LJ, Fan L, Xu M, Tang WL, Wright JM. Renin inhibitors vs angiotensin receptor blockers for primary hypertension. *Cochrane Database Syst Rev* 2025. <https://doi.org/10.1002/14651858.CD012570.pub2>.
- [30] Zetner D, Roost I, Rosenberg J, Andresen K. Prophylactic transarterial embolization in patients with bleeding peptic ulcers following endoscopic control of bleeding. *Cochrane Database Syst Rev* 2025. <https://doi.org/10.1002/14651858.CD014999.pub2>.

- [31] Samara MT, Kottmaier E, Helfer B, Leucht C, Christodoulou NG, Huhn M, et al. Switching antipsychotics vs continued current treatment in people with non-responsive schizophrenia. *Cochrane Database Syst Rev* 2025. <https://doi.org/10.1002/14651858.CD011885.pub2>.
- [32] Bader ER, Allam MM, Harris TG, Suchdev N, Loke YK, Barlas R. Thrombolysis for aneurysmal subarachnoid haemorrhage. *Cochrane Database Syst Rev* 2025. <https://doi.org/10.1002/14651858.CD013748.pub2>.
- [33] Wilting FN, Sondag L, Schreuder FH, Dammers R, Klijn CJ, Boogaarts HD. Surgery for spontaneous supratentorial intracerebral haemorrhage. *Cochrane Database Syst Rev* 2025. <https://doi.org/10.1002/14651858.CD015387.pub2>.
- [34] Otte WM, Vinkers CH, Habets PC, van IJzendoorn DGP, Tjebkink JK. Analysis of 567,758 randomized controlled trials published over 30 years reveals trends in phrases used to discuss results that do not reach statistical significance. *PLoS Biol* 2022;20:e3001562. <https://doi.org/10.1371/journal.pbio.3001562>.
- [35] Perneger TV. How to use likelihood ratios to interpret evidence from randomized trials. *J Clin Epidemiol* 2021;136:235–42. <https://doi.org/10.1016/j.jclinepi.2021.04.010>.
- [36] Wood J, Freemantle N, King M, Nazareth I. Trap of trends to statistical significance: likelihood of near significant *P* value becoming more significant with extra data. *BMJ* 2014;348:g2215. <https://doi.org/10.1136/bmj.g2215>.
- [37] Let's think about cognitive bias. *Nature* 2015;526:163. <https://doi.org/10.1038/526163a>.
- [38] Yao X, Xia J, Deodat M, Wang P, Chang Y, Luo Y, et al. GRADE notes 6: three strategies to determine the clinically important thresholds for outcomes in evidence-based guideline development. *J Clin Epidemiol* 2025;188:111976. <https://doi.org/10.1016/j.jclinepi.2025.111976>.
- [39] Royall R. On the probability of observing misleading statistical evidence. *J Am Stat Assoc* 2000;95:760–8. <https://doi.org/10.1080/01621459.2000.10474264>.